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NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2025
Odd Semester - 2025-26
HU102 - INDIAN KNOWLEDGE SYSTEM

Time: 3 Hrs

Maximum Marks : 60

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. Traditional dyes in ancient India were prepared from (1) C03 BL2 PO3
(a) Natural plants and minerals (b) Imported substances
(c) Chemicals (d) Petroleum products
2. Which ancient mathematician calculated the value of π accurately? (1) C02 BL1 PO2
(a) Brahmagupta (b) Patanjali
(c) Panini (d) Aryabhata
3. Who is known as the Father of Surgery in ancient India? (1) C04 BL1 PO4
(a) Charaka (b) Patanjali
(c) Dhanvantari (d) Sushruta
4. Buddhism belongs to which category of Indian philosophy? (1) C01 BL2 PO1
(a) Orthodox (b) Theistic
(c) Unorthodox (d) Ritualistic
5. What is the main advantage of decimal representation? (1) C02 BL2 PO2
(a) Decorative symbols (b) Only for trade
(c) Limited number usage (d) Easy calculation using place value
6. Indian Knowledge System mainly emphasizes (1) C01 BL1 PO1

- (a) Holistic and integrated knowledge (b) Industrial growth
- (c) Purely religious learning (d) Fragmented disciplines
7. Which of the following best reflects the unique aspect of IKS? (1) C01 BL1 PO11
- (a) Knowledge limited to written texts (b) Separation of science and philosophy
- (c) Integration of science, philosophy, art, and ethics (d) Dependence only on modern experimentation
8. What is the major historical contribution of Aryabhata to mathematics? (1) C02 BL2 PO1
- (a) Approximation of the value of π (b) Introduction of Roman numerals
- (c) Discovery of calculus (d) Invention of logarithms
9. Which ancient Indian advancement was crucial for water management?(1) C03 BL1 PO6
- (a) Cave paintings (b) Stepwells and tanks
- (c) Stone inscriptions (d) Palm-leaf manuscripts
10. Name one orthodox (Āstika) school of Indian philosophy. (1) C01 BL1 PO7
- (a) Buddhism (b) Jainism
- (c) Nyaya (d) Charvaka
11. The discovery of zero as a number is attributed to: (1) C02 BL1 PO1
- (a) Brahmagupta (b) Upagupta
- (c) Aryabhata (d) Bhaskara
12. The concept of zero was first clearly defined in (1) C01 BL1 PO1
- (a) China (b) Greece
- (c) Arabia (d) India

Part B**(Answer any four of the following) 5x4=20**

13. Name any two ancient Indian mathematicians and their contributions. (5) C02 BL1 PO2
14. Write a short note on the Vedangas. (5) C01 BL1 PO1
15. Write a note on shipbuilding in ancient India. (5) C03 BL1 PO3
16. Discuss traditional medicine and surgery in ancient India. (5) C04 BL4 PO4
17. Write a short note on Indian contributions to astronomy (5) C02 BL2 PO2

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| 18. Discuss mining and ore extraction practices in ancient India. | (5) | C03 | BL2 | PO3 |
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Part C
(Answer any three of the following) 3x10=30

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| 19. Explain the contributions of Aryabhata and Brahmagupta to mathematics and astronomy. | (10) | C02 | BL4 | PO2 |
| 20. Explain the role of traditional art forms and culture in preserving Indian knowledge. | (10) | C04 | BL3 | PO4 |
| 21. Describe dyes, painting traditions, and their significance in ancient India. | (10) | C03 | BL2 | PO3 |
| 22. Critically evaluate the contribution of ancient Indian metallurgy to global technological progress in the fields of weapons, construction, and fine arts. | (10) | C03 | BL5 | PO7 |
| 23. Describe the six orthodox (Astika) schools of Indian philosophy along with one key idea from each. | (10) | C01 | BL1 | PO3 |